INDIVIDUAL REPORT

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Q1

We are testing CAPM using the Fama-MacBeth Regression in the first task. With the help of the provided data, which includes 50 US equities, and the S&P 500 return as a stand-in for the market return, this method tests the validity of CAPM using the Fama-MacBeth 2-pass regression. The cross-sectional regression's intercept (α) should be statistically insignificant (around zero) if CAPM is correct. This suggests that there is no additional return unrelated to systematic risk after accounting for the impact of market beta (β_i). The applicability of the CAPM in explaining the excess returns of the 50 individual US equities depending on their market beta can be evaluated by running this Fama-MacBeth regression.

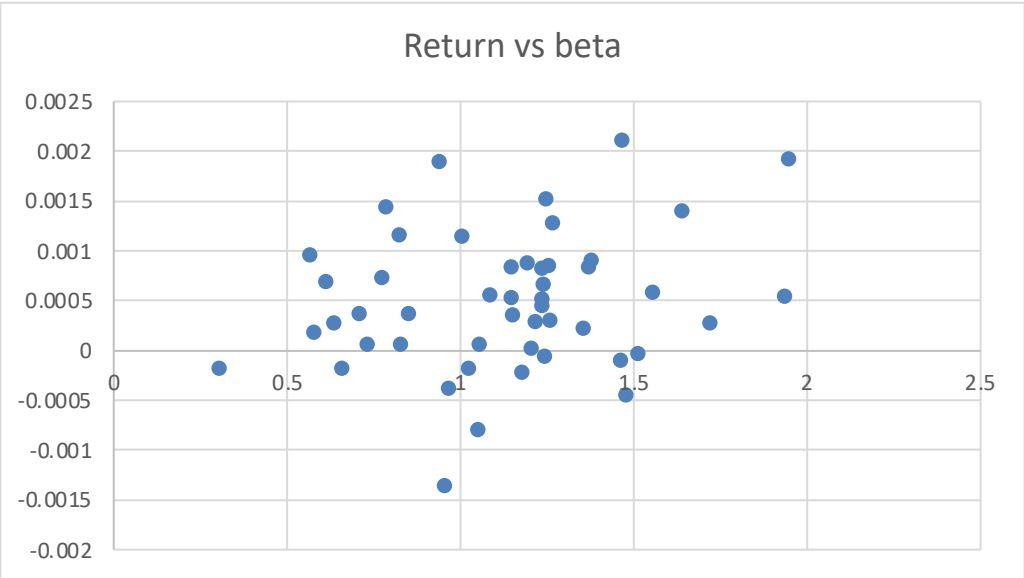
By fitting a straight line to the data, the statistical technique known as linear regression aims to model the connection between two continuous variables. This illustrates how steep the regression line is. When there is a positive slope, the y variable rises as the x variable rises, and vice versa when there is a negative slope. The y-value at which the regression line crosses the y-axis is known as the intercept. The table displays intercept values ranging from -0.00063 to 0.00145.This statistic shows how much of the variance in the y variable can be accounted for by the linear regression model. A higher number on the scale of 0 to 1 denotes a better fit.

Regression Statistics								
Multiple R	0.22370476							
R Square	0.05004382							
Adjusted R Sq	0.02983199							
Standard Erro	0.0006783							
Observations	49							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	1	1.1392E-06	1.1392E-06	2.47596645	0.12230666			
Residual	47	2.1624E-05	4.6009E-07					
Total	48	2.2764E-05						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	2.056E-06	0.00032812	0.00626587	0.99502711	-0.000658	0.00066215	-0.000658	0.00066215
X Variable 1	0.00043818	0.00027847	1.5735204	0.12230666	-0.000122	0.00099839	-0.000122	0.00099839

Slope and average are used in the regression model. The correlation between the independent and dependent variables is shown by this number, 0.2237. Higher numbers indicate a greater

link. The range is 0 to 1.The percentage of the dependent variable's variation that the model's independent variables can account for is shown by this value, 0.0500. A higher number on the scale of 0 to 1 denotes a better fit. In this instance, the variation is just 5% explained by the model. An adjusted R-square that takes the number of independent variables in the model into consideration is 0.0298. Models with varying amounts of independent variables are compared using it.

When all the independent variables are equal to zero, the intercept, 2.0566E-06, shows the expected value of the dependent variable. Keeping all other independent variables unchanged, the coefficient for the X variable, 0.000438, shows the change in the dependent variable that results from an increase of one unit in the independent variable. The statistical significance of this change is not shown by the p-value of 0.1223. This regression analysis's findings imply that the model does not adequately fit the data. The F-statistic (p-value = 0.1223) is not statistically significant, and the R-squared value is low (0.05). This indicates that a statistically significant amount of the variance in the dependent variable cannot be explained by the independent variables.



Q2

The regression intercept of the slope mean is used to get the expected return. The risk-free rate, variable intercept, and stock return for that specific date are used to compute the expected return. According to the table, the expected returns were constantly higher than the actual returns that were seen on that date. On the seventh date, the actual return (0.0070) exceeded the predicted return (0.00050285). On the 19th date, the actual return (0.0221) exceeded the predicted return (0.00050947). On the 43rd date, the actual return (0.0154) exceeded the predicted return (0.00050652). The actual may eventually surpass the predicted. All things considered, though, the projected returns appear to be consistently greater than the actual returns.

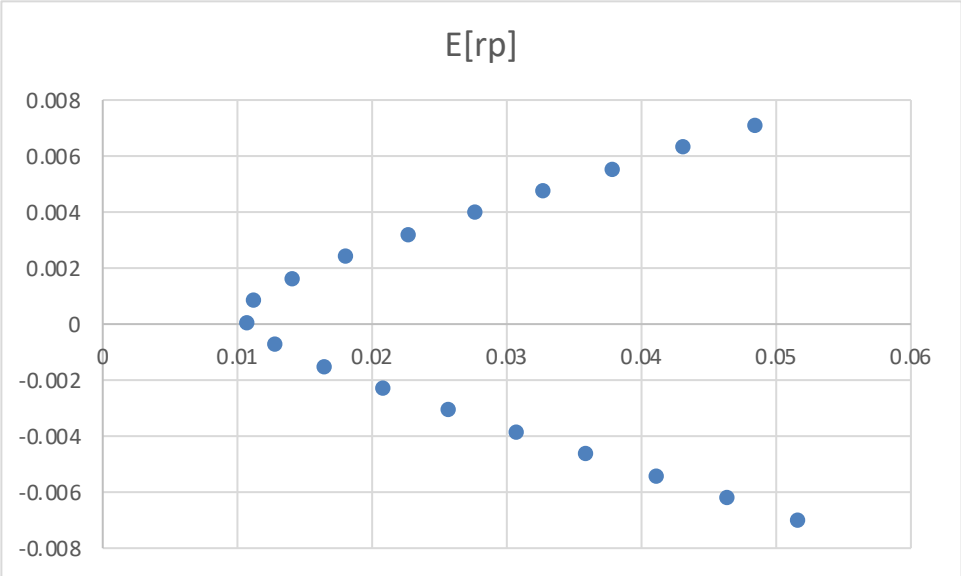
In event studies, statistical methods such as t-value and abnormal return (AR) are employed to evaluate the effect of an event on the price of a securities. Here, the t-value and abnormal return were utilized to indicate how significant of a difference in return we anticipated. The term "abnormal return" refers to the discrepancy between a security's actual return and its expected return during a given event window. A statistical test statistic called the T-Value is utilized to assess the statistical significance of the observed anomalous return.

The standard deviation of the abnormal returns determined during an event study is known as the STD of abnormal deviation. The variability (spread) of these aberrant returns for each security examined in the event study is captured by the SDAR. The abnormal return variability among securities is measured by the Standard Deviation of Abnormal Returns (SDAR). Here, the t-value was computed using SDAR. This allows us to determine the important value's significance.

In statistical hypothesis testing, the significance level, also represented by alpha (α), is an essential idea. It shows the likelihood of ruling out a valid null hypothesis. When it comes to the crucial value, the t-value is typically not significant in this portfolio we observed.

Q3

We calculate the risk return of 50 stocks in this area. We compute the expected return before calculating the excess return. The Variance Covariance Matrix was then calculated. Diversification, or distributing your investment over a variety of assets to lower overall risk, is a fundamental portfolio management concept. A foundational element of Modern Portfolio Theory (MPT), a paradigm for creating ideal portfolios, is the VCOV matrix. The VCOV matrix is a useful tool for determining the relative risk contribution of each asset in a portfolio. Value at Risk (VaR), a statistic that projects the possible maximum loss of a portfolio over a particular period with a given degree of confidence, requires the VCOV matrix as one of its fundamental inputs. To plot the efficient frontier, we calculated the expected return, variance, and standard deviation of two portfolios.Finally, we plotted the Efficient Frontier using various weights, and the C-shaped scatter chart is visible.

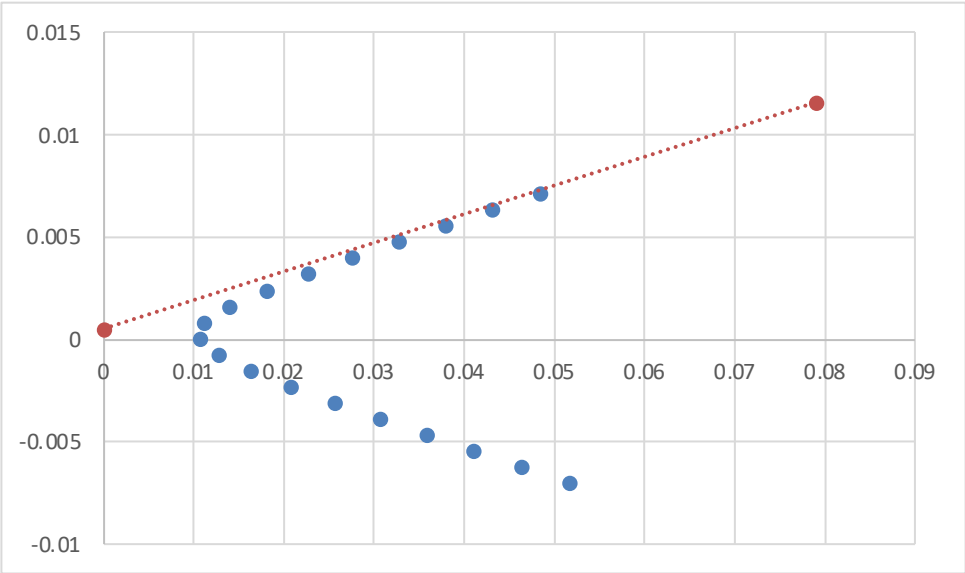


Q4

The weights of each stock that can be utilized for both long and short positions are assigned in this section. There are three limitations that need to be made for it. Stock should have a weight of no more than 20% and no less than -10%, with the total amount being equal to one. We utilize the solver function to determine the weights of the various stocks after constructing the Variance-Covariance matrix. To determine the expected return, variance, and standard deviation for each solver, we computed the Normal Solver both with and without constraint weightage. This will indicate the extent to which the weights and risk of the portfolios with and without constraints differ.

Q5

The Capital Market Line is plotted in the Efficient Frontier Line in this. In a market where there is a risk-free asset, investors can achieve an ideal balance between risk and return by following the Capital Market Line (CML). The CML is a visual that displays, for different portfolios, the expected return plotted against the risk (usually determined by standard deviation). It shows an upward-sloping straight line that represents a positive correlation between risk and return. Put more simply, you may expect larger profits if you take on more risk (higher standard deviation). The CML makes this ideal equilibrium easier to see. It recommends that investors look for portfolios that deliver the maximum projected return for a given degree of risk.



Q6

Hedging an Optimal Risky Asset Portfolio with Options

In certain cases, even after you've invested in a risk-adjusted portfolio with the best possible mix, you may still wish to reduce your downside risk. This is when having options is a useful tool. Here, we'll talk about utilizing options to hedge this portfolio and examine the reasoning behind the selected approach. By definition, an optimal portfolio seeks to maximize expected return for a specific level of risk tolerance. It is commonly built using Modern Portfolio Theory (MPT). By spreading your investment across a variety of assets, this "optimal mix" seeks to lower risk through diversification. Even a well-diversified portfolio, meanwhile, is susceptible to unanticipated market downturns. Diversification reduces risk, but it doesn't completely remove it. Unforeseen incidents or market movements may result in large losses. By establishing a floor price for your portfolio, hedging assists in reducing your potential losses. Even if the ideal portfolio is built for long-term gain, it might be unsettling to experience short-term market volatility. During these times, hedging might provide brief protection. Having a hedge in place can give you peace of mind, particularly in times of market volatility (TEAM, 2023).

Hedging Strategy: Protective Puts

The best options strategy is to deploy protective puts, since the purpose of an ideal portfolio is to minimize downside risk. You purchase put options on an index or exchange-traded fund (ETF) that closely mirrors the performance of your ideal portfolio. With these put options, you have the option—but not the duty—to sell the underlying asset (index/ETF) by the expiration date at a fixed price (the strike price). For the benefit of selling at the strike price, you must pay a premium up advance. The price of hedging is reflected in this premium. You can exercise your right to sell at the strike price of the put options if the market value of your portfolio drops below that price, so limiting your losses (KENTON, 2021). The total amount of loss you can incur is the original put option premium plus any possible drop in the portfolio's value above the strike price. It is not required of you to use the put options. You only lose the premium if the market rebounds before the options expire and become worthless. Based on your risk tolerance and the state of the market, you can adjust the level of protection by selecting the put options' strike price and expiration date. The strike price,

expiration date, and present volatility of the market all affect the cost of put options. A larger premium is associated with more protection. The value of your put options will decline in the event of a large market advance, which will reduce your potential upside gains in comparison to an unhedged portfolio. Using options as a hedge calls for proactive management. It is up to you to keep an eye on the market and determine when to buy and sell put options (Anon., 2024).

Protective puts can be used to limit downside risk while keeping exposure to possible growth in an ideal portfolio. When you desire peace of mind during market turbulence or have a short-term need for protection, this technique can be quite helpful. Before putting such a plan into action, it's important to take the expenses and possible restrictions of possibilities into account. It's crucial to keep in mind that options are sophisticated financial products and employing them should only be done after consulting a financial expert.

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